

comes to power applications, higher power efficiency is always the guiding factor for new technology." Power of electric motors can range anywhere from milli-watts to mega-watts. "Intelligent performance and improved efficiency of both motor and driver circuitry are the areas of design focus," he adds.

When it comes to electronics driving the motor, innovation in the fields of topology, power devices and integration are defining the roadmap. Anand adds, "One of the critical elements for these systems is power sources with the capability to deliver higher current in peaks."

Integrated solutions

Solutions that consume lower power (around/less than 50W) are now available with integrated power devices and state machine to control. For example, DRV10983, a three-phase sensor-less motor controller with integrated power MOSFETs, can provide continuous drive current up to 2A.

With integrated-state machine, driver and power MOSFETs, it uses a proprietary sensor-less control scheme to provide continuous sinusoidal drive, which reduces pure-tone acoustics that typically occur as a result of commutation. Interface to the device is designed to be simple and flexible. EEPROM is integrated in DRV10983 as memory for the motor parameter and operation settings.

DRV10970 is an integrated three-phase BLDC motor driver for home appliances, cooling fans and other general-purpose motor-control applications. Embedded intelligence, small form factor and simple pin-out structure help with design complex-

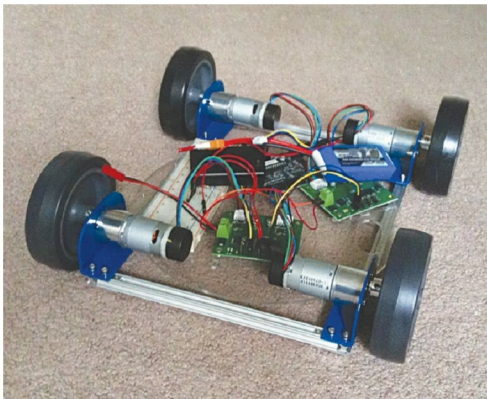


Fig. 2: A motor in application

ity, board space and system cost. Integrated protections can keep a watchful eye on system robustness and reliability.

TC78S122FNG is a low-power-consuming module with up to four channels for motor with DC brush. Application areas include surveillance cameras, banking terminals, video slot machines and home appliances like refrigerators and air-conditioners. Currently available in DIP package, Toshiba also plans to bring out a quad-flat package very soon.

Features to look forward to

Kilby Labs engineers have developed and patented a sensor-less control algorithm that turns power electronics into smart gate drivers and eliminates the need for a position sensor. The end result is

a more reliable motor system that will start a motor every time under full-load conditions. The system also continues to drive and control the motor as load and speed performance change, as required by the application.

Kamat says, "We now have a very wide variety to select from." As we move ahead, improving efficiency will always be the focus. Anand adds, "Minimising power losses is a key objective of designs." Gallium-nitride, silicon-carbide and other such newer semiconductor materials might enable in reaching higher efficiencies.

Total harmonic distortion is another concern in motor-driving applications. Currently-used passive filtering mitigates this to some effect, but since it is bulky, it can introduce load based resonance. Combining active and passive methods (hybrid) or pure active methods could, however, be feasible solutions to increase system performance. These technologies are being researched at some research and development centres. **EFY**

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